Collatz Covering Claim in Lean 4: Type-Checked Formal Specification of the Open 2-Adic Cylinder Covering Conjecture

Benjamin Stanley Frohman
ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/collatz-covering-claim
Dedicated 1-to-1 Zenodo DOI: 10.5281/zenodo.23653007

Earliest Commit Provenance: 2026-10-04 03:46:42 PM CDT (UTC-5)
Siloed Archival Edition: October 25, 2026 (America/Chicago CDT)

Abstract

Standalone Lean 4 formalization (‘CoveringClaim.lean‘, ‘CoveringClaim/Claim.lean‘, ‘CoveringClaim/Defs.lean‘) giving a clean, type-checked formal specification of the open Collatz covering claim over residue classes modulo powers of two.

1 Scope, Scientific Honesty, and Non-Claim Boundary

Formalizes the exact mathematical statement of the open Collatz covering claim in Lean 4 without claiming a proof of the Collatz conjecture.

- **Repository:** `https://github.com/BenFrohman/collatz-covering-claim`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23653007`
- **Earliest Git Commit SHA:** `ffa8cbc068a873f8c3917eaaf33de8c7f466fb83` (2026-10-04 03:46:42 PM CDT (UTC-5))
- **License & Copyright:** Copyright © 2026 Benjamin Stanley Frohman. Licensed under the Apache License, Version 2.0.

2 Technical Overview & Repository Specification

Copyright (c) 2026 Benjamin Stanley Frohman.
License: Apache-2.0
This repository contains one equation: every positive integer reaches 1.
“‘lean
def RemainingLemma : Prop :=
n : Nat, 0 < n → k : Nat, iter k n = 1
axiom remaining_lemma_assumption : RemainingLemma
“‘
That axiom is not a proof. The Terras density equation is not in this repository. It lives in ‘collatz-terras-density‘. A first descent is not arrival at 1.

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/collatz-covering-claim` is audited via `titan-audit` (`audit/TITAN_AUDIT_CERTIFICATE.json`) and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
`CoveringClaim/Claim.lean`	847 B	`64ad431fe1b0d794...`
`CoveringClaim.lean`	227 B	`b86592e32d0d9a02...`
`LICENSE`	11355 B	`9da72c2bee4f20ba...`
`NOTICE`	266 B	`1f0e025930a15c71...`
`README.md`	1045 B	`5e6d0ec5c31215f7...`
`lake-manifest.json`	127 B	`134de28c4d13cf6a...`
`lakefile.lean`	294 B	`4b5594733e7d2761...`
`lean-toolchain`	25 B	`8733782dc070a99b...`

Attribution & Licensing Notice: Copyright © 2026 Benjamin Stanley Frohman (ORCID: `0009-0006-7068-3718`). Licensed under the Apache License, Version 2.0.